

BOND DURATION & CONVEXITY

[A] BOND DURATION

I) What is Bond Duration:

Bond duration measures how sensitive a bond's price is to interest rate changes, expressed in years. The core rule is inverse: if interest rates rise by 1%, a bond's price will fall by a percentage roughly equal to its duration (and vice versa).

Key Concepts

- **Macaulay Duration:** The weighted-average time it takes to receive all of the bond's cash flows (both coupons and principal), calculated in years.
- **Modified Duration:** Adjusts Macaulay duration to directly estimate the percentage change in a bond's price for a given change in yield to maturity (YTM).
- **Effective Duration:** Used for bonds with embedded options (like callable bonds), calculating price sensitivity based on fluctuating interest rates.

How Factors Affect Duration

- **Time to Maturity:** Bonds that take longer to mature have higher durations, making them more sensitive to interest rate risk.
- **Coupon Rate:** Bonds with higher coupon payments return your invested capital faster, leading to a lower duration and reduced interest rate sensitivity.

II) Are Higher Duration Bonds Better or Worse?

Higher duration bonds are not inherently "better"; they simply carry more **interest-rate risk**. A bond's duration measures how much its price changes when interest rates shift. Higher duration means greater volatility.

- **When they are better:** Higher duration bonds (or long-term bonds) are better if you expect **interest rates to fall**. Because bond prices rise when rates drop, long-duration bonds will gain much more value. They also offer **higher yields** as a reward for tying up your money.
- **When they are worse:** If **interest rates rise**, high-duration bonds are much worse. For example, if rates increase by 1%, a bond with a 10-year duration will drop roughly 10% in price.
- **Rule of thumb:** Imagine **duration as a seesaw**. When the seesaw of interest rates goes up, the price of your high-duration bond goes way down. When rates go down, your price goes way up.

[B] BOND CONVEXITY

III) What is Bond Convexity?

Bond convexity measures the curvature in the relationship between bond prices and interest rates. While duration estimates this relationship as a straight line, true price movements form a curve. It helps investors gauge a bond's price sensitivity and correct the inaccuracies of standard duration calculations.

Key Characteristics of Convexity

- **Price Movement:** Bonds with positive convexity appreciate more when interest rates fall than they depreciate when rates rise by the same margin.
- **Determinants:** Convexity increases with longer maturities and lower coupon rates. Zero-coupon bonds offer some of the highest convexity because all cash flows occur at the end.
- **Positive vs. Negative Convexity:**
 - **Positive Convexity:** Standard option-free bonds (like most government and standard corporate bonds) exhibit positive convexity, working in the investor's favour.
 - **Negative Convexity:** Bonds with embedded options, such as callable bonds or mortgage-backed securities, often display negative convexity. As interest rates drop, the bond might be called or refinanced, capping your price gains while keeping your downside risk intact.

IV) Are Higher Convexity Bonds Better or Worse?

Higher bond convexity is generally better. It means a bond's price will rise more when interest rates fall and drop less when rates rise. This asymmetry favours the investor. However, these bonds usually offer lower yields, meaning you pay upfront for this protection.

Here is how higher convexity works:

- **The Curve of the Bond:** Bond duration is a basic tool that guesses how bond prices change. It acts like a straight line. Convexity is the true curve of this relationship.
- **When Rates Drop:** Imagine a seesaw. When interest rate falls, bond prices go up. A high convexity bond curves upward. It gains extra value beyond what a straight-line duration would predict.
- **When Rates Rise:** When rates go up, bond prices drop. A high convexity bond has a curve that flattens out. It loses less value than expected.

- **The Trade-Off:** Bonds with the highest convexity (like zero coupon bonds) have lower interest payments. You essentially pay a higher price for this "insurance" against rate changes

[C] HOW DURATION & CONVEXITY MIGHT IMPACT BONDS DUE TO INTEREST RATE CHANGES

V) Impact of Duration and Convexity in an Environment of Rate Hikes

In an environment of policy rate hikes by the central bank, a high-duration, high-positive-convexity bond will experience a significant drop in price, but the losses will be noticeably smaller than what a standard duration estimate would predict.

The interaction between these two bond metrics during rate hikes operates as follows:

1. The Impact of High Duration

- **Concept:** Duration measures how sensitive a bond's price is to interest rate changes.
- **Effect:** A high duration means the bond's price is highly volatile.
- **The Result:** When the central bank hikes policy rates, the bond will suffer a severe price decline.

2. The Benefit of High Positive Convexity

- **Concept:** Convexity measures the "curvature" of the bond price/yield relationship. Standard duration assumes this relationship is a straight line, but it is actually a curved line.
- **Effect:** Positive convexity means the bond's price rises faster when rates fall and drops less when rates rise.
- **The Result:** The high positive convexity acts as a natural buffer or shock absorber. Even though the bond loses value, the "curved" nature of the price change minimizes the damage.

(VI) Impact of Duration and Convexity in an Environment of Rate Cuts

A combination of a high duration and high positive convexity bond will result in a massive capital appreciation when the central bank cuts interest rates. The bond's high duration makes its price very sensitive to falling yields, while its positive convexity supercharges those gains by curving the price upward.

These two forces interact during rate cuts in a few specific ways:

- **Massive Price Gains (Duration Effect):** Because bond duration measures how sensitive a bond's price is to interest rate movements, a higher duration means larger percentage price increases when interest rates drop.

- **Curved Upside (Convexity Effect):** Bond convexity measures how much a bond's price changes beyond what duration alone predicts. Positive convexity means that as yields drop, the bond's price increases at a faster and faster rate, amplifying your returns.
- Zero-coupon bonds are a classic example of this. Because they do not pay regular interest, all their value is realized at maturity, leading to very high duration and maximum positive convexity. If the central bank aggressively cuts policy rates, the price of these long-term zero-coupon bonds will skyrocket, resulting in significant profits for the investor.